

Glomerular (Nephrotic) — Minimal Change Disease



1. 70-90% kids

WHAT YOU SEE

Stone gate carved '70-90% KIDS / 10-15% ADULTS'

WHAT IT ENCODES

Minimal change disease (MCD) is the #1 cause of nephrotic syndrome in CHILDREN, accounting for ~70–90% of pediatric nephrotic syndrome (peak age ~2–6 years). In adults it causes only ~10–15% of nephrotic syndrome (membranous and FSGS dominate adults). Presentation is classic abrupt nephrotic syndrome: heavy proteinuria, hypoalbuminemia, edema, hyperlipidemia, often after a viral URI.

BOARD TRAP

WRONG ANSWER: 'FSGS is the most common cause of nephrotic syndrome in children.' Discriminator: in CHILDREN, MCD is by far #1 (70–90%); FSGS leads in Black adults and is a more common pediatric cause only when MCD is steroid-resistant. For a young child with sudden nephrotic syndrome, the answer is MCD, treated empirically with steroids without biopsy.

2. LM normal

WHAT YOU SEE

Woman at microscope, sign 'LM NORMAL'

WHAT IT ENCODES

Light microscopy is essentially NORMAL — hence 'minimal change.' Glomeruli look unremarkable with no hypercellularity, no sclerosis, and no thickened basement membrane; the only abnormality is on electron microscopy. This normal LM with full nephrotic syndrome is the defining paradox of MCD.

BOARD TRAP

WRONG ANSWER: 'segmental sclerosis (scarred glomerular segments) is seen on light microscopy.' Discriminator: segmental sclerosis on LM defines FSGS, not MCD — MCD has NORMAL light microscopy. Because FSGS is focal/segmental, a biopsy that samples only spared glomeruli can look 'normal' and mimic MCD; true steroid-resistance should prompt re-biopsy to find missed FSGS.

3. Foot process effacement

WHAT YOU SEE

'FOOT PROCESS EFFACEMENT' with EM magnifier over child

WHAT IT ENCODES

The sole diagnostic abnormality is on electron microscopy: DIFFUSE effacement (flattening/fusion) of podocyte foot processes from podocyte injury, with intact basement membrane and no deposits. The effacement is reversible, which is why proteinuria resolves rapidly with steroids. MCD is the prototypical podocytopathy.

BOARD TRAP

WRONG ANSWER: 'subepithelial spikes / immune-complex deposits on EM.' Discriminator: subepithelial deposits with 'spike-and-dome' = membranous nephropathy, not MCD. MCD shows ONLY diffuse foot process effacement with NO electron-dense deposits — any deposit on EM excludes pure MCD.

4. EM only

WHAT YOU SEE

Magnifying glass labeled 'EM ONLY'

WHAT IT ENCODES

Because light microscopy and immunofluorescence are unrevealing, ELECTRON MICROSCOPY is required to make a definitive histologic diagnosis (showing foot process effacement). In practice, however, biopsy is usually deferred in children because the diagnosis is presumed clinically and treated empirically.

BOARD TRAP

WRONG ANSWER: 'MCD is diagnosed by light microscopy alone.' Discriminator: LM is normal in MCD, so it cannot make the diagnosis — the abnormality (foot process effacement) is visible only on EM. The bigger clinical point: in a typical child you don't biopsy at all and treat empirically with steroids.

5. No immune deposits

WHAT YOU SEE

Slit-pore membrane / 'NIL LESION' on immunofluorescence

WHAT IT ENCODES

Immunofluorescence is NEGATIVE — there are no immune-complex deposits (no IgG, no C3), which is why MCD is also called 'nil lesion' disease. The proteinuria is from loss of the podocyte charge barrier, not immune-complex-mediated injury, distinguishing it mechanistically from membranous and proliferative GN.

BOARD TRAP

WRONG ANSWER: 'granular IgG/C3 deposits along the capillary loops on immunofluorescence.' Discriminator: granular subepithelial IgG/C3 ('granular' or 'spike') = membranous nephropathy; MCD has NEGATIVE IF (nil lesion). Any positive immune staining argues against MCD.

6. Selective proteinuria

WHAT YOU SEE

Bucket 'SELECTIVE PROTEINURIA' with albumin balloons

WHAT IT ENCODES

MCD causes SELECTIVE proteinuria — predominantly albumin is lost because the injury is loss of the glomerular negative charge (anionic) barrier, which preferentially leaks small, negatively charged albumin while larger proteins (IgG) are retained. Selective albuminuria correlates with steroid responsiveness.

BOARD TRAP

WRONG ANSWER: 'non-selective proteinuria with loss of albumin AND larger plasma proteins (IgG, etc.).' Discriminator: non-selective proteinuria (size-barrier loss) is typical of FSGS/membranous and more advanced glomerular damage; MCD is characteristically SELECTIVE (mostly albumin) due to pure charge-barrier loss. Non-selective proteinuria argues against MCD.

7. Heavy proteinuria >3.5g

WHAT YOU SEE

Cup '10 g/24h' / '3 g/24h'

WHAT IT ENCODES

Proteinuria is nephrotic-range, >3.5 g/24h, and can be massive (10 g/day or more), producing the full nephrotic syndrome (hypoalbuminemia, edema, hyperlipidemia). Despite the heavy protein loss, hematuria, hypertension, and renal insufficiency are typically ABSENT in classic MCD — a 'pure' nephrotic picture.

BOARD TRAP

WRONG ANSWER: 'sub-nephrotic proteinuria <1 g/day with active urine sediment.' Discriminator: MCD produces HEAVY nephrotic-range proteinuria (>3.5 g/day) with a BLAND sediment — sub-nephrotic protein with hematuria/casts points to a nephritic process, not MCD. Prominent hematuria/HTN/azotemia should make you doubt classic MCD and consider FSGS or a nephritic GN.

8. Steroid responsive

WHAT YOU SEE

Trophy 'STEROIDS 1ST LINE / 8 WEEKS / 90% KIDS / 80% ADULTS'

WHAT IT ENCODES

MCD is the most steroid-RESPONSIVE glomerular disease: prednisone (children ~2 mg/kg/day or 60 mg/m²/day; adults ~1 mg/kg/day) induces remission in ~90% of children and ~80% of adults, usually within weeks (children often respond by ~4 weeks; adults can take up to 8–16 weeks). High steroid responsiveness is itself a diagnostic clue.

BOARD TRAP

WRONG ANSWER: 'MCD is steroid-resistant and requires cytotoxic agents as first-line therapy.' Discriminator: MCD is the textbook STEROID-RESPONSIVE nephrotic disease — corticosteroids are first line and most patients remit. Cytotoxic/second-line agents are reserved for frequent relapsers or steroid dependence, not initial therapy; true steroid-RESISTANCE should make you reconsider FSGS.

9. No biopsy first

WHAT YOU SEE

Sign 'NO BIOPSY FIRST'

WHAT IT ENCODES

In a typical child (age ~1–10) with classic nephrotic syndrome, MCD is so likely that you treat EMPIRICALLY with corticosteroids WITHOUT a kidney biopsy; biopsy is reserved for atypical features (age <1 or >10–12, gross hematuria, hypertension, low complement, renal insufficiency) or steroid resistance. Adults, by contrast, usually DO get a biopsy because the differential is broader.

BOARD TRAP

WRONG ANSWER: 'always biopsy a child before starting treatment for nephrotic syndrome.' Discriminator: standard pediatric practice is empiric steroids first, biopsy only for atypical presentations or steroid resistance — routine upfront biopsy is wrong in the classic child. (The opposite applies to most adults, where biopsy is standard.)

10. Relapse / merry-go-round

WHAT YOU SEE

Merry-go-round 'RELAPSE / FREQUENT RELAPSE / STEROID DEPENDENT'

WHAT IT ENCODES

Although most patients respond, relapses are common: many are 'frequent relapsers' or 'steroid-dependent' (relapse on tapering or shortly after stopping steroids). Repeated/prolonged steroid courses cause cumulative toxicity (growth suppression, cushingoid features, osteoporosis), which is what drives the move to steroid-sparing second-line agents.

BOARD TRAP

WRONG ANSWER: 'a single course of steroids is always curative in MCD.' Discriminator: MCD is highly steroid-RESPONSIVE but frequently RELAPSING — a meaningful fraction become frequent relapsers or steroid-dependent and need steroid-sparing therapy. Equating 'responsive' with 'permanently cured' is the trap.

11. Secondary: NSAIDs/Hodgkin's/lithium

WHAT YOU SEE

'ASSOCIATIONS' sign — owl (Hodgkin's), NSAIDs, lithium pills

WHAT IT ENCODES

Secondary/associated MCD: NSAIDs (can cause MCD often with concurrent acute interstitial nephritis), Hodgkin lymphoma (the classic paraneoplastic association — think MCD as a paraneoplastic phenomenon, via aberrant T-cell cytokines), and lithium. Identifying and removing the trigger (stop the drug, treat the lymphoma) is part of management.

BOARD TRAP

WRONG ANSWER: 'hepatitis B is the classic association of MCD.' Discriminator: hepatitis B is associated with MEMBRANOUS nephropathy (and HBV/HCV with MPGN), NOT MCD. MCD's signature associations are NSAIDs, Hodgkin lymphoma, and lithium. Matching the secondary cause to the right pattern is the testable point.

12. Atopy 40%

WHAT YOU SEE

Boy sneezing, 'ATOPY 40%'

WHAT IT ENCODES

MCD is associated with atopy/allergic disease in roughly a third to 40% of cases (eczema, asthma, allergic rhinitis, recent allergen exposure), consistent with the proposed T-cell/cytokine (immune dysregulation) mechanism; onset is often triggered by a viral URI or allergic stimulus.

BOARD TRAP

WRONG ANSWER: 'MCD is associated with chronic infection (e.g., chronic hepatitis or endocarditis-type immune-complex disease).' Discriminator: chronic infection-driven, immune-complex nephropathies (membranous/MPGN) are a different mechanism; MCD links to ATOPY/allergy and T-cell dysregulation, with negative immunofluorescence (nil lesion). Don't attach an immune-complex/chronic-infection story to MCD.

13. Steroid-resistant → biopsy / FSGS

WHAT YOU SEE

Laptop 'STEROID RESISTANT? DIG DEEPER' near 'FSGS'

WHAT IT ENCODES

Failure to respond to an adequate steroid course (steroid resistance) should prompt a kidney biopsy and reconsideration of the diagnosis — most often it reveals FSGS, which can be missed on a limited biopsy because its lesions are focal/segmental. Steroid responsiveness vs resistance is a key fork distinguishing MCD from FSGS.

BOARD TRAP

WRONG ANSWER: 'persistent steroid resistance still represents classic MCD — just give more steroids.' Discriminator: classic MCD is steroid-responsive, so true steroid resistance argues AGAINST MCD and for FSGS (or genetic podocytopathy) — the move is to biopsy and re-evaluate, not to escalate steroids indefinitely.

14. 2nd line agents

WHAT YOU SEE

'2ND LINE' — cyclophosphamide, calcineurin inhibitor, MMF, rituximab

WHAT IT ENCODES

For frequently relapsing or steroid-dependent MCD, steroid-sparing second-line agents are used: cyclophosphamide, calcineurin inhibitors (cyclosporine/tacrolimus), mycophenolate mofetil (MMF), and rituximab (increasingly favored, especially in relapsing pediatric disease). These reduce cumulative steroid toxicity while maintaining remission.

BOARD TRAP

WRONG ANSWER: 'escalate straight to dialysis/transplant for relapsing MCD.' Discriminator: MCD almost never progresses to ESKD — relapsing/steroid-dependent disease is managed with steroid-sparing immunosuppression (cyclophosphamide, CNI, MMF, rituximab), not dialysis. Jumping to renal replacement therapy for MCD is wrong because the long-term renal prognosis is excellent.
